

PLS-SEM: A method demonstration in the R statistical environment

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Abstract

In line with calls to stimulate methodological diversity and support evidence-based human resource development (HRD) through quantitative competencies, we present a methods demonstration leveraging open-source tools and lesser-known quantitative research methods to support the HRD research community and applied HRD in the workplace. In this paper, we provide an informative introduction to partial least squares structural equation modeling (PLS-SEM). We discuss PLS-SEM application trends in the field of HRD, present key characteristics of the method, and demonstrate up-to-date metrics and evaluation guidelines using an illustrative model. Our PLS-SEM demonstration and explanations can serve as a valuable resource for practitioners concerned with substantiating results for organizational stakeholders and support researchers in methodological decision-making while avoiding common pitfalls associated with less familiar methods. Our step-by-step demonstration is conducted in open-source software and accompanied by explicitly coded operations so that readers can easily replicate the illustrative analyses presented.

KEYWORDS

causal-predictive modeling, HRD, human resource development, mediation analysis, partial least squares structural equation modeling, PLS-SEM, R statistical software, SEMinR

Structural equation modeling (SEM) is a family of statistical procedures that allow researchers to model and simultaneously analyze multiple independent and dependent constructs combined with the ability to distinguish between

observed and latent (unobserved) variables (Kline, 2016). As a generalized framework, SEM experienced rapid diffusion and advancement over the past 50 years—and we have only seen the beginning of its evolutionary journey (Bollen et al., 2022). Researchers and scholar-practitioners concerned with human behavior often find themselves confronted with highly complex realities of the people and systems they aim to analyze and understand. For this reason, SEM is widely utilized by researchers in the social, behavioral, management, and decision sciences (Deng et al., 2018; Tarka, 2017). Within the context of human resource development (HRD) research, SEM is not only a common quantitative approach for assessing interventions and outcomes (Shirmohammadi et al., 2020), but a predominant quantitative method employed by HRD researchers (Park et al., 2021).

As the SEM methodology gained momentum and diffused across disciplines throughout the 20th and 21st centuries, it developed new characteristics, special cases, and diverse applications. One noteworthy development being nonparametric SEM methods (Bollen et al., 2022). The present study focuses on a nonparametric variation of the SEM framework known as partial least squares SEM (PLS-SEM; Chin, 1998; Lohmöller, 1989; Wold, 1982). Traditional, covariance-based SEM (CB-SEM) applications (typically referred to simply as SEM in HRD discourse) have a longer history of use across social and behavioral sciences, including HRD. As a causal-explanatory method, the primary goal of CB-SEM is “to confirm (or reject) theories and their underlying hypotheses” (Hair et al., 2021, p. 4), or in other words, “to test whether the theoretical model is supported by sample data” (Schumacker & Lomax, 2016, p. 1). PLS-SEM was introduced as an alternative causal-predictive approach to SEM (Jöreskog & Wold, 1982, p. 270), which focuses on explaining the variance in the model's dependent variables (Chin et al., 2020). Thus, as described by Sarstedt, Ringle, and Hair (2022), the identification of key success factors and sources of competitive advantage is an important selection criterion for the application of PLS-SEM.

Despite the reported absence of PLS-SEM applications in HRD research (Legate et al., 2023), the method has experienced substantial growth in applied business and management settings for more than a decade. In fact, the prevalence of PLS-SEM across a broad range of social sciences has been marked by exponentially increasing trends in both publications and citations (see Figure 1).

Where it appears that PLS-SEM has developed into a useful method for research in a variety of business contexts, its value within the field of HRD remains to be seen. Just as the SEM methodological framework has evolved,

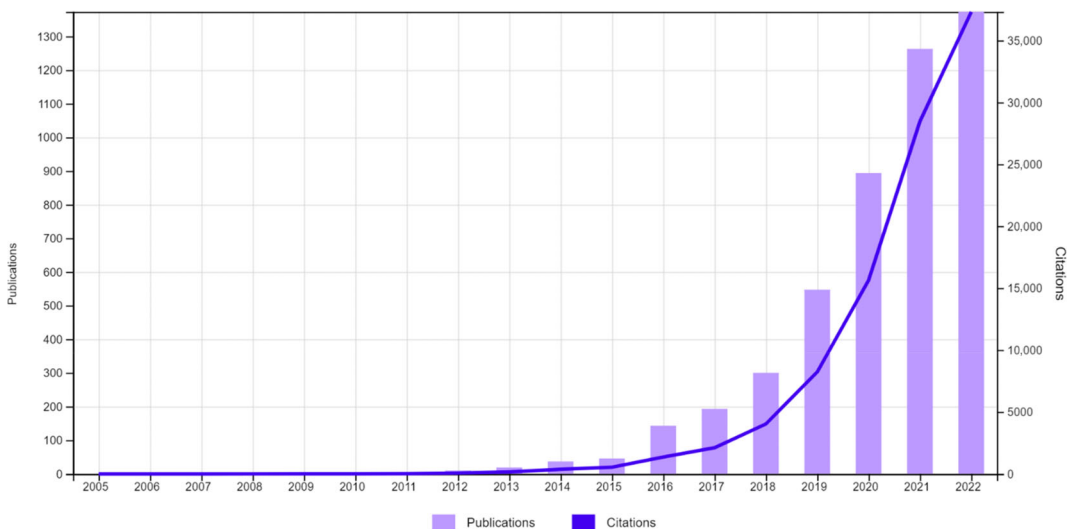


FIGURE 1 PLS-SEM times cited and publications over time (Web of Science). Results from Social Sciences Citation Index (SSCI). Source: Publications selected from Web of Science Core Collection (as of December 2022). Search terms: TS = partial least squares structural equation model* or TS = “PLS-SEM” and publication years 2005–2022.

so too have HRD practices, organizations, and workforces, inspiring calls for the adoption of diverse investigation methods to keep pace with the increasingly complex nature of work (Park et al., 2021; Short & Gray, 2018; Tkachenko et al., 2017). Similarly, proponents of evidence-based HRD research direct attention to a pressing need for developing analytical research competencies among HRD professionals (Gubbins et al., 2018; Shirmohammadi et al., 2020). Gubbins et al. (2018), for example, emphasized quantitative competencies related to identifying, assessing, and analyzing evidence-based findings as key to establishing trust between HRD professionals and stakeholders (p. 197). Yet, engaging in applied research to inform evidence-based decision-making necessitates not only access to the best available information (Gubbins et al., 2018) and appropriate statistical software, but the application of rigorous methodologies and validated assessment metrics (Aldrich et al., 2015).

In line with calls to stimulate methodological diversity and to support evidence-based HRD through quantitative competencies, our purpose is twofold. First, we aim to provide a general understanding of the PLS-SEM method, associated metrics, and evaluation guidelines among HRD scholars and practitioners. This paper can serve as an informative introduction to PLS-SEM, especially for those within the HRD community who conduct research and/or collaborate with researchers in related business and management fields where PLS-SEM is increasingly applied. The methodological guidance offered in this article can also assist researchers in deciding which SEM approach is appropriate for their intended application while avoiding common pitfalls associated with lesser-known methods. Second, we aim to support evidence-based HRD by advancing quantitative research competencies in a variation of SEM developed for theory-soft, data-heavy organizational contexts (Wold, 1982). Our PLS-SEM demonstration and explanations can serve as a valuable resource for HRD practitioners charged with planning interventions, assessing outcomes, and substantiating results for organizational stakeholders. Furthermore, we present the PLS-SEM demonstration and corresponding code using open-source software so researchers can easily replicate presented analyses and modify code for future use, without financial or contractual commitments.

In this paper, we describe characteristics of PLS-SEM and explain features that may be of interest to HRD researchers. Although we describe PLS-SEM as it compares with CB-SEM throughout, explanations are purely intended to highlight features of PLS-SEM from a familiar frame of reference (i.e., CB-SEM). We do not view them as competing methods, but rather, as unique quantitative research tools available to researchers within the evolving, unified SEM framework (Bollen et al., 2022). Advantages of one method often represent disadvantages of another, and vice versa (Dash & Paul, 2021; Hair, Hult, et al., 2017; Jöreskog & Wold, 1982; Rigdon et al., 2017). Unlike PLS-SEM, refereed CB-SEM applications in Association for Human Resource Development (AHRD) sponsored journals offer a wealth of guidance for readers seeking to learn about accepted methodological justifications and up-to-date evaluation metrics (see Legate et al., 2023; Park et al., 2021; Shirmohammadi et al., 2020). Therefore, in the present study, we emphasize situations in which PLS-SEM may be useful to HRD researchers taking into account current guidance for its rigorous application.

Nevertheless, criticism of PLS-SEM has been a topic of discourse in prior research (e.g., Henseler et al., 2014; Marcoulides et al., 2012; Petter, 2018; Russo & Stol, 2023; Sarstedt et al., 2016; Sharma, Liengaard, Sarstedt, et al., 2023). A full review of PLS-SEM debates is beyond the scope of the present study. We do, however, encourage readers to review and consider varying perspectives to ensure responsible application of the PLS-SEM method (Hair et al., 2019; Petter & Hadavi, 2021). For readers interested in side-by-side comparisons, Rigdon et al. (2017, p. 5) reviewed a substantial collection of studies comparing the efficacy of PLS-SEM and CB-SEM and summarized results in a reader-friendly table. In a recent article published in the *Journal of Business Research*, Cook and Forzani (2023) presented a balanced, detailed synopsis of relevant discourse across social sciences literature. The authors elaborated on criticisms and evolving disagreements in the “PLS-SEM versus CB-SEM debate” (Cook & Forzani, 2023, p. 1) which prepares readers with an understanding of perspectives not covered in the present study. Readers may also benefit from reviewing tab. 1.6 in Hair et al. (2021, chap. 1, p. 22) for rules of thumb when choosing between CB-SEM and PLS-SEM.

The remainder of the paper is structured as follows. First, we present evidence surrounding the application of PLS-SEM in HRD research. Second, we provide an overview of characteristics of PLS-SEM which may prove useful

to the HRD community and discuss some of the method's limitations. We then describe the illustrative model and discuss key research design and analytic considerations for our demonstration. This is followed by a guided, step-by-step demonstration of PLS-SEM using the SEMinR package (Ray et al., 2022) developed for the R statistical software (R Core Team, 2022). We conclude with a discussion of implications for HRD and practice, study limitations, and suggestions for future research.

1 | PLS-SEM IN HRD RESEARCH

The PLS-SEM path modeling approach was developed in the early 1980s (Jöreskog & Wold, 1982; Lohmöller, 1989; Wold, 1982), but was only recently introduced to the *Human Resource Development Quarterly* (HRDQ) readership as a complementary SEM approach for obtaining prediction-oriented solutions by Legate et al. (2023) who demonstrated execution of PLS-SEM using the statistical software SmartPLS (Ringle et al., 2015). Upon reviewing past issues of HRDQ to examine the extent to which PLS-SEM had been applied within the HRDQ community, Legate et al. (2023, pp. 2–3) found that while SEM was the predominant quantitative method employed (i.e., 37%), all cases applied CB-SEM. Similar results were reported by Park et al. (2021), p. 8) who captured empirical research trends across select HRD journals from 2016 to 2020; findings likewise evidenced the predominant application of SEM (41%). As the latter did not differentiate between SEM variations, it remains undetermined the extent to which PLS-SEM has been applied by the HRD research community.

To expand this line of inquiry, we reviewed a broader range of HRD journals to identify applications of PLS-SEM. As in prior studies, our review included journals sponsored by the AHRD as well as the *European Journal of Training and Development* (EJTD; see Pham-Duc et al., 2023; Shirmohammadi et al., 2020). We conducted a search using the University of Texas at Tyler Robert R. Muntz Library online database collection on the search terms (“HRD” OR “Human Resource Development”) AND [“PLS” OR “PLS-SEM” OR “Partial Least Squares”]. Nine unique PLS-SEM articles were identified. Results included six articles published in EJTD and one article each published in *Advances in Developing Human Resources*, *Human Resource Development International*, and HRDQ. Eight of the nine studies reported original empirical research, the remaining article was the aforementioned methods paper (i.e., Legate et al., 2023). While these results are non-exhaustive and too few to draw meaningful conclusions, they offer preliminary evidence of PLS-SEM as a nascent methodology in the HRD community. Throughout the following section, we incorporate examples from these empirical studies to link our discussion points to HRD research contexts.

2 | PLS-SEM: FLEXIBLE MODELING FOR COMPLEX REALITIES

In this section, we discuss systematic ways in which PLS-SEM differs from CB-SEM through the lens of flexibility for HRD research in complex organizational settings. Our summary of concepts is necessarily high-level throughout due to the paper's introductory nature and intended audience. Interested readers are referred to Bollen et al. (2022) for an exhaustive history of the SEM framework and to Sarstedt et al. (2023) for discussion of ongoing debates and critiques that have spurred advances and enhanced methodological rigor in PLS-SEM. Finally, Shmueli (2010) provided an accessible overview to aid researchers in distinguishing predictive and explanatory statistical modeling.

2.1 | To predict or explain?

Researchers in fields such as human resource management (Yuan et al., 2021) and psychology (Yarkoni & Westfall, 2017) have been exploring the benefits of predictive modeling techniques to capture unknown phenomena and contribute to theory building. Causal-explanatory modeling, which represents the majority of HRD quantitative studies (Park et al., 2021), involves the application of statistical models for the purpose of testing hypotheses about

underlying structures in covariance matrices (Cliff, 1983). More plainly, the researcher's focus is on testing theoretically specified associations in observational data for which the causality is implied through established theory (Hair et al., 2021; Shmueli, 2010). Alternatively, in causal-predictive modeling, researchers apply predictive statistical models or algorithms to model observations with the goal of predicting new data or future observations (Shmueli, 2010; Shmueli et al., 2016; Yarkoni & Westfall, 2017). Regression models are examples of traditional predictive modeling approaches (Yuan et al., 2021). The predictive modeling process is appropriate for exploring and capturing relationships that are theoretically less formalized or previously unknown as it enables researchers to engage with models in “a more free-form fashion” (Shmueli, 2010, p. 290).

In HRD, researchers have applied PLS-SEM to capture predictive drivers of professional returnees' adaption to the workplace (Kunasegaran et al., 2016), explore organizational factors that drive employees' protean career orientation (Wong et al., 2017), and evaluate how incivility in the workplace impacts work stress and knowledge sharing among public sector employees (Alias et al., 2022). As a causal-predictive method (Chin et al., 2020; Jöreskog & Wold, 1982), PLS-SEM's algorithm maximizes explained variance in dependent constructs and observed indicators (Lohmöller, 1989; Manley et al., 2021; Wold, 1982). Thus, in the most simplistic form, PLS-SEM emphasizes (in-sample) prediction when estimating statistical models (Sarstedt, Hair, et al., 2022). In addition to metrics associated with in-sample prediction and explanatory characteristics (e.g., R^2 and f^2), recent PLS-SEM methodological developments offer researchers sophisticated procedures and metrics for assessing a given model's out-of-sample prediction (Liengaard et al., 2021; Sharma et al., 2023; Shmueli et al., 2016, 2019). These expanded analytical techniques enable practitioners to strategically complement organizational analyses with metrics developed for the prediction of new data (Hair, 2021; Hair & Sarstedt, 2021; Sarstedt & Danks, 2021). In our PLS-SEM demonstration, we introduce readers to procedures and metrics for assessing out-of-sample prediction (i.e., $PLS_{predict}$) within the context of PLS-SEM.

2.2 | Complexity meets narrowly defined populations

HRD researchers in business settings are faced with the reality that conducting research in real businesses can be messy, uncertain, and constrained by feasibility considerations (Bryman & Bell, 2015; Guenther et al., 2023). Flexible traits underlying the PLS-SEM method (Jöreskog & Wold, 1982; Lohmöller, 1989; Wold, 1982) can support HRD researchers in investigations targeting narrowly defined populations, such as small business research where methodological choices are limited due to the nature of the population (Short & Gray, 2018). CB-SEM is considered a large-sample technique in that it requires larger sample sizes with increased model complexity (i.e., more parameters; Kline, 2016, pp. 14–15). Plainly stated, sampling considerations are more impactful in CB-SEM, especially in cases of non-normal data distribution (Jannoo et al., 2014; Kahn, 2006). As a nonparametric method, PLS-SEM imposes weak assumptions about population distributions. This characteristic, however, is not a silver bullet (Sarstedt et al., 2023) as it imposes certain challenges for evaluating results (e.g., bootstrapping; Becker et al., 2023; Rigdon et al., 2017).

Nonetheless, these flexible features, combined with an algorithm that iteratively estimates the model through a series of ordinary least squares (OLS) regressions, results in a path modeling process that enables researchers to maintain necessary levels of scale and model complexity when exceedingly large samples are not attainable (Jöreskog & Wold, 1982). While all analytical techniques benefit from larger samples, in some cases the nature of the population justifies the use of alternative methods, like PLS-SEM, which can accommodate a broader range of sample sizes for applied research (Kock & Hadaya, 2018; Rigdon, 2016). This was the case for HRD researchers, Mustafa and Sim (2023), who targeted a narrowly defined population of employee-supervisor dyads ($n = 136$). According to the authors, due to the exploratory nature of their research the study required specification of a complex model and PLS-SEM was selected as the method of choice (Mustafa & Sim, 2023).

2.3 | From simple to sophisticated modeling procedures

The HRD empirical studies we identified in our search included not only complex model constellations, but more sophisticated special cases of SEM. These included hierarchical component models (Alias et al., 2022), conditional mediation models (Mustafa & Sim, 2023; Wong & Roziah, 2015), and multigroup analysis (Wong & Roziah, 2019). To elaborate on an example, researchers often desire to examine mediation (i.e., direct and indirect effects) and moderation (i.e., interaction terms) within the context of a SEM path model (Hair et al., 2022). PLS-SEM facilitates both mediation (Memon et al., 2018; Nitzl et al., 2016) and moderation testing (e.g., Becker et al., 2023; Memon et al., 2019) within the modeling process, without requiring large sample sizes or ad hoc analyses (Sarstedt, Hair, et al., 2020). In causal models applying CB-SEM, assessment of conditional mediation is regularly achieved by means of ad hoc regression analyses (Hayes & Rockwood, 2020). In HRD, for example, both Wong and Roziah (2015, $n = 306$) and Mohd Rasdi et al. (2023; $n = 273$), elected PLS-SEM based on a predictive research focus and facilitation of conditional mediation without the need to conduct separate regression analyses.

Another special case of SEM is multigroup analysis, which can arise when research goals necessitate accounting for natural groupings in populations of interest (e.g., departments, demographic characteristics, etc.; Yuan et al., 2021). For example, HRD researchers Wong and Roziah (2019, $n = 188$) applied PLS-SEM multigroup analysis (i.e., PLS-MGA) to examine predictors of self-directed career orientation and assess whether gender had an impact on these associations. Another flexible feature of PLS-SEM that has been leveraged by HRD researchers is formative measurement models for latent, multi-item constructs (Hair et al., 2022). Formatively measured constructs are less commonly modeled in HRD research (i.e., reflectively measured constructs are typical; see Legate et al., 2023) but are easily incorporated into simple or complex PLS-SEM applications, thereby increasing the ability of structural models to accommodate alternative causal (formative) data measurement composites (Sarstedt et al., 2016).

An example of a formatively measured construct is socioeconomic status. As a latent variable, socioeconomic status is typically measured by income, education, and occupational status—in this case a researcher would not suggest that these indicators are *caused* by an individual's socioeconomic status, but rather, they form the construct (Russo & Stol, 2023). Full explanation of formative measurement concepts is beyond the scope of the present study; for additional information, we refer readers to Coltman et al. (2008), Diamantopoulos et al. (2008), Diamantopoulos and Winklhofer (2001), and Sarstedt et al. (2016). Among the HRD PLS-SEM studies identified, Kunasegaran et al. (2016) operationalized workplace adaptation as a formatively measured three-dimensional construct comprised of job knowledge, acculturation, and productive work relationships. While the PLS-SEM studies identified by our search are limited and represent a small portion of advanced PLS-SEM applications available to researchers, they do present evidence of diversity and innovation in quantitative research methods within the HRD research community.

2.4 | Limitations and cautions

Introducing readers to opportunities and extant guidance for rigorous application of PLS-SEM is a goal of this paper, but it is not our intention to promote PLS-SEM as a panacea for all HRD research problems. As a referee pointed out, PLS-SEM is not a master key to causal, multi-level problems that should be used by HRD researchers. Proposing that HRD researchers should use PLS-SEM would be as misguided as suggesting that SEM is a statistical cure for all organizational research problems. On the contrary, regardless of method, no single study examining a specific phenomenon can provide sufficient evidence of an effect or offer the scientific rigor necessary to establish theory (Rigdon et al., 2017; Russo & Stol, 2023). Where PLS-SEM may prove useful for some HRD research questions, it does have limitations and is not a silver bullet to solve all the problems that scientists face in their empirical research (Hair et al., 2011; Marcoulides & Saunders, 2006; Sarstedt et al., 2023; Sosik et al., 2009). For example, CB-SEM is appropriate for research designs which focus on testing and confirming concise theoretical models based on global goodness of fit measures and which necessitate models that consider, for example, correlations or reverse relationships in the structural model or correlated error

terms in the measurement models (Hair et al., 2022). PLS-SEM relies on quantitative data, and therefore, is subject to data quality standards. No statistical method has the magical properties required to transform poor data and inadequate sampling strategy into sound inferences for large populations (Petter, 2018). While PLS-SEM accommodates a range of sample sizes where populations are constrained, disregard for sampling theory guidelines will lead to questionable results. For example, exceedingly small samples and extremely nonnormal data increase risk of Type II errors in PLS-SEM. Likewise, researchers conducting PLS-SEM must pay close attention to missing data, influential outliers, and collinearity which can influence the underlying OLS regressions or inflate standard errors from bootstrapping (Hair et al., 2022). In the following sections, we provide guidance for consideration when conducting PLS-SEM analyses.

3 | A PLS-SEM DEMONSTRATION

To introduce the application of the PLS-SEM method, we present an example of a simple PLS path model that includes a mediator. We show with this example how researchers and practitioners who are less familiar with PLS-SEM can establish their model, estimate it with PLS-SEM, assess the results using appropriate criteria, and report them in a comprehensible way. We first introduce the R Statistical Environment (R Core Team, 2022), followed by an overview of the model used for this demonstration including key elements of the illustrative research design. We then provide a step-by-step demonstration of PLS-SEM model assessment and mediation analysis using the SEMinR package (Ray et al., 2022). The R code and dataset are available in the Open Science Framework repository (Legate & Ringle, 2022).

3.1 | The R statistical environment

Existing software applications support a broad range of SEM applications. For PLS-SEM, researchers can execute analyses in commercial PLS-SEM software with graphical user interface such as SmartPLS (Ringle et al., 2022) or WarpPLS (Kock, 2022). Extant software reviews by Memon et al. (2021) and Sarstedt and Cheah (2019) offer more information for researchers seeking user-friendly, commercial PLS-SEM software options. In this paper, we demonstrate how to execute PLS-SEM model assessment and mediation testing using the SEMinR package (Ray et al., 2022) developed for the open source R statistical environment (R Core Team, 2022). While some users prefer the intuitiveness of a user-friendly interface and the ease of executing advanced analyses offered by commercial software, script-based software also has advantages. First, the transparency of explicitly coded operations better facilitates replication and can lend confidence to research findings (Russo & Stol, 2023, p. 129). Second, given limited applications of PLS-SEM in HRD for researchers to draw upon for guidance, practitioners may have little motivation to commit financial resources to subscription-based software.

While tech-savvy researchers may readily promote R as an accessible tool for learners, others note that becoming familiar with R is like learning a new language (Hackenberger, 2020). For this reason, throughout each stage of the demonstration, we provide detailed descriptions of commands along with full syntax (i.e., code) and output to support researchers new to the R statistical environment. As we will demonstrate, a computer programming background is not required for utilizing R for applied business research. Our demonstration closely follows the recently published, open access workbook on PLS-SEM in R by Hair et al. (2021). Furthermore, all supplementary materials supporting the illustrative model are housed in a public repository so that readers can easily reproduce all demonstrated analyses (Legate & Ringle, 2022).

3.2 | The illustrative commitment-performance model

We build on a prior HRDQ method demonstration (see Legate et al., 2023) by examining the illustrative commitment-performance model (CPM) adapted from Hair et al. (2020). The Conceptual Model of Workplace Commitment developed by Fornes et al. (2008, p. 347) serves as the theoretical framework underpinning this research

design (Creswell & Creswell, 2018). Based on an integrative review of literature and concept mapping, Fornes et al. (2008) developed this framework as a practical guide for operationalizing HRD interventions which are proposed to lead to enhanced workplace commitment, and ultimately, improved performance. The demonstrated analyses are based on cross-sectional survey research and for this demonstration, we utilize a partially simulated dataset. The analytic approach, PLS-SEM, is deemed appropriate for this illustrative study because our goal is to explore the most impactful predictors of the target performance outcome to better inform strategic planning for organizational performance improvement (Hair et al., 2022).

3.2.1 | Modeled constructs

The presented model constellation includes four constructs (i.e., variables) of interest. Exogenous (also referred to as “predictor” or “driver”) variables include supervision and teamwork. The theoretical CPM model is depicted in Figure 2.

Based on Fornes et al. (2008), each of the predictors is expected to impact employee performance directly and indirectly through employees' level of organizational commitment (the mediator). Figure 3 provides a graphical representation of each CPM construct from a measurement model perspective. As displayed in Figure 3, supervision (SUPR), teamwork (TEAM), and organizational commitment (ORGC) are reflectively measured with three indicators each based on employee self-reported survey data. The key outcome, performance (PERF), is measured using a single-item indicator based on employee ratings by immediate supervisors. Table 1 contains an overview of relevant labeling, measurement, and item response information for the CPM.

3.2.2 | Sample size

To determine the technically required minimum sample size for PLS-SEM, we follow the inverse square root approach proposed by Kock and Hadaya (2018). Assuming the minimum path coefficient expected to be significant will be 0.1/0.2/0.3, approximately 618/155/69 observations are needed for a corresponding effect significant at 5% (Kock & Hadaya, 2018; see also Hair et al., 2021, chap. 1, p. 15).

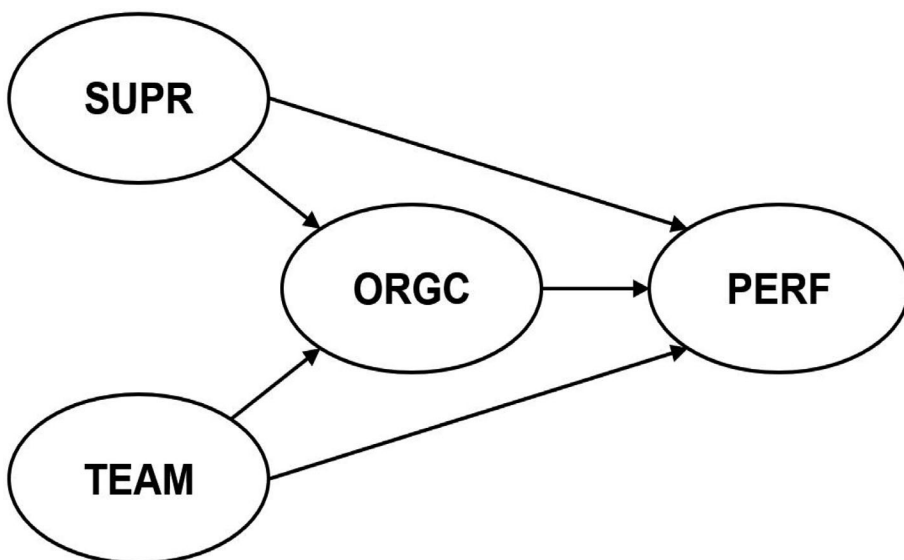


FIGURE 2 Theoretical model. Theoretical model for methodological demonstration purposes. Adapted from Hair et al. (2020). Copyright 2020 by Taylor & Francis Group. Reproduced by permission of Taylor & Francis Group.

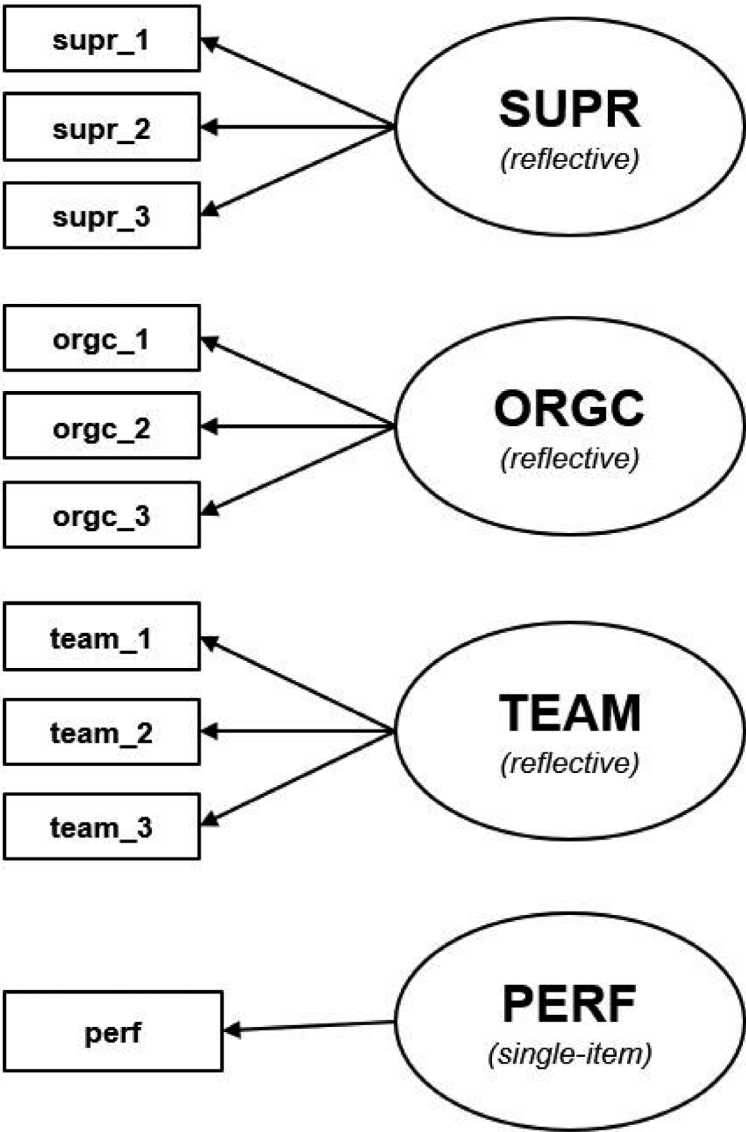


FIGURE 3 Measurement models. All constructs are reflectively measured. ORGC, organizational commitment; PERF, performance; SUPR, supervision; TEAM, teamwork.

The sample size for this example is 91. The standard formulas for calculating sample size based on the inverse square root approach (Kock & Hadaya, 2018) are presented in Figure 4. The minimum sample size ($n_{\min}$) is given by these formulas, respectively, where $p_{\min}$ is the absolute value of the minimum path coefficient (see Figure 4; see also Guenther et al., 2023).

3.3 | Demonstration setup

To guide readers through the analyses, we follow the same presentation format throughout all stages of the demonstration. Each section is supported with a table containing the R code and corresponding output (e.g., Table 2). Each section of code is presented in a gray box which also contains a brief description. Descriptive text is preceded by the

TABLE 1 Commitment-performance model construct measurement details.

Construct	Label	Measurement model	Data source	Scale	Sample item
Supervision	SUPR	Reflective	Self-Report Survey Data	3 items; 7-point scale (1 = <i>Strongly disagree</i> , 7 = <i>Strongly agree</i>)	“My supervisor is friendly and helpful.”
Teamwork	TEAM	Reflective	Self-Report Survey Data	3 items; 7-point scale (1 = <i>Strongly disagree</i> , 7 = <i>Strongly agree</i>)	“My immediate work group functions as a team.”
Organizational commitment	ORGC	Reflective	Self-Report Survey Data	3 items; 7-point scale (1 = <i>Strongly disagree</i> , 7 = <i>Strongly agree</i>)	“I have a sense of loyalty to this company.”
Employee performance	PERF	Single indicator	Supervisor Survey Data	Single item; 7-point scale (1 = <i>Very low performance</i> ; 7 = <i>Very high performance</i>)	Employees were rated by their supervisor on a 7-point scale.

Note: Adapted from Hair et al. (2020). Copyright 2020 by Taylor & Francis Group.

Significance level = 1%	Significance level = 5%	Significance level = 10%
$n_{\min} > \left(\frac{3.168}{ p_{\min} } \right)^2$	$n_{\min} > \left(\frac{2.486}{ p_{\min} } \right)^2$	$n_{\min} > \left(\frac{2.123}{ p_{\min} } \right)^2$

FIGURE 4 Formulas for calculating sample minimum sample size. Formulas for calculating minimum required sample size for PLS-SEM based on the inverse square root approach proposed by Kock and Hadaya (2018).

symbol “#,” which tells R to ignore that statement when running the section of code. The output generated by R is presented in a white box immediately following its corresponding code.

3.3.1 | Installing R and accessing data files

To conduct the step-by-step analyses, users will first need to download, install, and run R (<https://www.R-project.org>) or RStudio, the integrated development environment for R (<https://www.rstudio.com/>). The R syntax file and dataset (available from <https://osf.io/jhcx4/>) can be opened and used in either R or RStudio.

3.3.2 | Installing SEMinR package and loading in the dataset

This preliminary section details how to initiate the project, install the SEMinR package, and load in the CPM dataset (Table 2). Before conducting an R analysis, we recommend creating a project folder on your device to store all related files, including the R syntax (i.e., code) and the underlying dataset. You will establish this folder as your working directory at the beginning of the R session using the `setwd()` command. To begin analyzing the CPM, open a new R project, then install the SEMinR package (Ray et al., 2022) via `install.packages()` and load the SEMinR library using the `library()` function. You will need to load any required packages and set your working directory at the

TABLE 2 Initial steps for CPM in R.

Code 2.1	# DOWNLOAD & INSTALL SEMinR PACKAGE install.packages("seminr")
Code 2.2	#LOAD SEMinR LIBRARY library(seminr)
Code 2.3	#SET WORKING DIRECTORY setwd("C:\\Users\\....")
Code 2.4	#LOAD THE CPM DATASET CPM <- read.csv("CPM_Data.csv", header = TRUE, sep = ";")
Code 2.5	#VIEW OBSERVATIONS nrow(CPM)
Output 2.5	[1] 91

Note: The data should be prepared in a comma-separated value (CSV) file. The first row contains the variable names (i.e., headers) and the subsequent rows are the observations. To import your own data, prepare the data matrix following this format in Excel, and save as a CSV file.

beginning of each R session and every time you restart R. Next, load in the CPM dataset from your working directory using the `read.csv()` function. When we load the “CPM.csv” file, we need to indicate that this dataset has headers and uses a semicolon as a data separator. To view the number of observations in our dataset, we use the `nrow()` function. This dataset has no missing values. Otherwise, we would indicate the coding of the missing values (e.g., `missing_value = “-99”`). You can use the `?` operator to find help with a function in R (e.g., `?read.csv`) at any time during the session.

3.4 | Model specification and estimation

For recommended guidelines on conducting PLS-SEM using the SEMinR package, we refer to the open access PLS-SEM in R book (Hair et al., 2021). To begin our CPM demonstration, we first specify each of the measurement models and then the path relationships in the structural model (Table 3). SEMinR uses the `constructs()` function to specify the list of all construct measurement models. The `composite()` function is used to describe the measurement model of each specific construct in the model. Hair et al. (2021, chap. 4) explain in detail the arguments of the `composite()` function. SEMinR further distinguishes between `multi_items()` and `single_item()` composites (Table 3, Code 3.1). When no measurement weighting scheme is specified, the argument default is set to `mode_A`. Alternatively, researchers can use a `mode_B` specification for the composite's estimation. Researchers usually use `mode_A` for the estimation of reflectively conceptualized constructs, while they use `mode_B` for the estimation of formatively conceptualized constructs (Sarstedt et al., 2016). For more details on specifying reflectively and formatively measured constructs in the SEMinR package, see Hair et al. (2021, chap. 3, p. 58). To specify the structural model, we use the `relationships()` function to assign the relationships between all constructs and `paths()` to specify specific relationships between sets of antecedents and outcomes. The arguments `from` and `to` are applied within the path function to establish the direction of these relationships. After specifying the modeled relationships for the CPM, we will estimate the measurement and structural models. In SEMinR, we use the `estimate_pls()` function to estimate the PLS-SEM model. Once the model has been estimated, we use `summary()` to extract output and parameters of importance from our estimated model. The summary results will then be contained within the `summary_CPM_model` object we call upon multiple times throughout the demonstration.

TABLE 3 Specify and estimate the model.

Code 3.1	<pre>#CREATE MEASUREMENT MODELS pls_mm <- constructs(composite("SUPR", multi_items("supr_", 1:3)), composite("TEAM", multi_items("team_", 1:3)), composite("ORGC", multi_items("orgc_", 1:3)), composite("PERF", single_item("perf")))</pre>
Code 3.2	<pre>#CREATE STRUCTURAL MODEL pls_sm <- relationships(paths(from = c("SUPR","TEAM"), to = c("ORGC")), paths(from = c("SUPR","TEAM","ORGC"), to = c("PERF")))</pre>
Code 3.3	<pre>#MODEL ESTIMATION CPM_model <- estimate_pls(data = CPM, measurement_model = pls_mm, structural_model = pls_sm,)</pre>
Output 3.3	Generating the semirn model All 91 observations are valid.
Code 3.4	<pre>#SUMMARIZE MODEL RESULTS summary_CPM_model <- summary(CPM_model)</pre>

3.5 | Measurement model evaluation

At this stage, the model has been specified and estimated and we now have a summary of model output and parameters. Before we can evaluate the CPM structural (i.e., “inner”) model, we first need to assess the quality of our measurement (i.e., “outer”) models. In this demonstration, we follow procedures for evaluating reflective models as outlined by Hair et al. (2021, chap. 4) and Sarstedt, Ringle, and Hair (2022). For each multiple-item measurement model (SUPR, TEAM, and ORGC), we will review (a) indicator reliability, (b) internal consistency reliability, (c) convergent validity, and (d) discriminant validity. Indicator reliability is assessed by reviewing the outer loadings (minimum threshold ≥ 0.708 ; see also Legate et al., 2023). For internal consistency reliability we examine Cronbach's alpha (conservative), composite reliability rhoC values (liberal), and reliability coefficient rhoA, which typically falls between the lower-bound alpha and upper-bound rhoC values (Sarstedt et al., 2023). The internal consistency reliability should range between 0.60 and 0.95, whereby results ≥ 0.60 are considered satisfactory for exploratory research, outcomes ≥ 0.70 are good, and results ≥ 0.95 may be problematic in that excessively high associations between indicators suggests redundancy (e.g., Hair et al., 2021, chap. 4, p. 77). Convergent validity is established based on a minimum average variance extracted (AVE) of 0.50. The heterotrait–monotrait ratio (HTMT) of correlations (Henseler et al., 2015; Ringle et al., 2023) is applied to assess discriminant validity. HTMT results exceeding 0.85 (or 0.90 for conceptually similar constructs) are indicative of discriminant validity problems when modeled constructs are conceptually distinct (Henseler et al., 2015).

Table 4 provides code and output for reflective measurement model assessment. The `summary_CPM_model` object (created in Table 3, Code 3.4) enables us to extract and inspect the indicator loadings (`$loadings`) element (Table 4, Code. 4.1). To review measures for composite reliability and convergent validity, we inspect the `$reliability` elements within the `summary_CPM_model` object (Code 4.2). Finally, the HTMT is assessed in SEMinR by inspecting the `validity` element for the `$htmt` (Code 4.3).

Based on the output (see Table 4, Output 4.1 through 4.3), measurement model (i.e., SUPR, TEAM, ORGC) results are within acceptable ranges for outer loadings (≥ 0.708), composite reliability ($\alpha \geq 0.819$, rhoC = 0.885–

TABLE 4 Evaluate measurement model(s).

Code 4.1	#INSPECT OUTER LOADINGS summary_CPM_model\$loadings																																																							
Output 4.1	<table><tr><td></td><td>SUPR</td><td>TEAM</td><td>ORGC</td><td>PERF</td></tr><tr><td>supr_1</td><td>0.931</td><td>0.000</td><td>0.000</td><td>0.000</td></tr><tr><td>supr_2</td><td>0.924</td><td>0.000</td><td>0.000</td><td>0.000</td></tr><tr><td>supr_3</td><td>0.807</td><td>0.000</td><td>0.000</td><td>0.000</td></tr><tr><td>team_1</td><td>0.000</td><td>0.899</td><td>0.000</td><td>0.000</td></tr><tr><td>team_2</td><td>-0.000</td><td>0.790</td><td>0.000</td><td>0.000</td></tr><tr><td>team_3</td><td>0.000</td><td>0.854</td><td>0.000</td><td>0.000</td></tr><tr><td>orgc_1</td><td>0.000</td><td>0.000</td><td>0.908</td><td>0.000</td></tr><tr><td>orgc_2</td><td>0.000</td><td>0.000</td><td>0.898</td><td>0.000</td></tr><tr><td>orgc_3</td><td>0.000</td><td>0.000</td><td>0.883</td><td>0.000</td></tr><tr><td>perf</td><td>0.000</td><td>0.000</td><td>0.000</td><td>1.000</td></tr></table>		SUPR	TEAM	ORGC	PERF	supr_1	0.931	0.000	0.000	0.000	supr_2	0.924	0.000	0.000	0.000	supr_3	0.807	0.000	0.000	0.000	team_1	0.000	0.899	0.000	0.000	team_2	-0.000	0.790	0.000	0.000	team_3	0.000	0.854	0.000	0.000	orgc_1	0.000	0.000	0.908	0.000	orgc_2	0.000	0.000	0.898	0.000	orgc_3	0.000	0.000	0.883	0.000	perf	0.000	0.000	0.000	1.000
	SUPR	TEAM	ORGC	PERF																																																				
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orgc_3	0.000	0.000	0.883	0.000																																																				
perf	0.000	0.000	0.000	1.000																																																				
Code 4.2	#INSPECT COMPOSITE RELIABILITY AND CONVERGENT VALIDITY summary_CPM_model\$reliability																																																							
Output 4.2	<table><tr><td></td><td>alpha</td><td>rhoC</td><td>AVE</td><td>rhoA</td></tr><tr><td>SUPR</td><td>0.866</td><td>0.919</td><td>0.791</td><td>0.865</td></tr><tr><td>TEAM</td><td>0.819</td><td>0.885</td><td>0.721</td><td>0.878</td></tr><tr><td>ORGC</td><td>0.878</td><td>0.925</td><td>0.804</td><td>0.885</td></tr><tr><td>PERF</td><td>1.000</td><td>1.000</td><td>1.000</td><td>1.000</td></tr></table> Alpha, rhoC, and rhoA should exceed 0.7 while AVE should exceed 0.5		alpha	rhoC	AVE	rhoA	SUPR	0.866	0.919	0.791	0.865	TEAM	0.819	0.885	0.721	0.878	ORGC	0.878	0.925	0.804	0.885	PERF	1.000	1.000	1.000	1.000																														
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PERF	1.000	1.000	1.000	1.000																																																				
Code 4.3	#INSPECT THE HTMT CRITERION summary_CPM_model\$validity\$htmt																																																							
Output 4.3	<table><tr><td></td><td>SUPR</td><td>TEAM</td><td>ORGC</td><td>PERF</td></tr><tr><td>SUPR</td><td>.</td><td>.</td><td>.</td><td>.</td></tr><tr><td>TEAM</td><td>0.220</td><td>.</td><td>.</td><td>.</td></tr><tr><td>ORGC</td><td>0.503</td><td>0.521</td><td>.</td><td>.</td></tr><tr><td>PERF</td><td>0.392</td><td>0.535</td><td>0.816</td><td>.</td></tr></table>		SUPR	TEAM	ORGC	PERF	SUPR	.	.	.	.	TEAM	0.220	.	.	.	ORGC	0.503	0.521	.	.	PERF	0.392	0.535	0.816	.																														
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0.925, rhoA = 0.865–0.885), convergent validity (AVE ≥ 0.721), and discriminant validity (HTMT ≤ 0.816). Thus, we retain all items and proceed to evaluation of the structural model.

3.6 | Structural model evaluation

Initial steps for evaluation of the structural model include assessment of (a) collinearity, (b) significance and relevance of paths, and (c) explanatory power (Hair et al., 2021, chap. 6; Sarstedt, Ringle, & Hair, 2022). Collinearity is assessed via the variance inflation factor (VIF). VIF values of 3.0 or higher are indicative of the likelihood of collinearity issues in the model. In PLS-SEM, the relevance and significance of the structural paths is assessed via the size of the path coefficients and bootstrapping. The bootstrapping procedure (e.g., based on 10,000 subsamples) estimates standard errors used to compute confidence intervals (CIs) for significance testing. For assessing the statistical significance, the estimated *t*-values should exceed 1.960 (Hair et al., 2021, chap. 6).

To assess explanatory power, we will review *R*² values (i.e., variance explained by model predictors), which should be sufficiently high in comparison with results from similar studies. Effect sizes (*f*²) are expected to reflect the rank order of the path coefficients (Hair et al., 2021, chap. 6). As detailed in Table 5 (Code 5.1), we begin by evaluating collinearity of predictor constructs in relation to each endogenous construct by examining `$vif_antecedents` in the `summary_CPM_model`. We use the `bootstrap_model()` function based on 10,000 subsamples (`nboot=10000`) to bootstrap our previously estimated SEMinR model (Code 5.2). Note that `alpha` sets the specified level for significance. Next, the bootstrapping results are accessed by inspecting the `bootstrapped_paths`

TABLE 5 Inspect the structural model.

Code 5.1	#INSPECT STRUCTURAL MODEL COLLINEARITY (VIF) summary_CPM_model\$vif_antecedents																																																							
Output 5.1	ORGC : SUPR TEAM 1.017 1.017 PERF : SUPR TEAM ORGC 1.261 1.379 1.683																																																							
Code 5.2	#BOOTSTRAP THE MODEL boot_CPM_model <- bootstrap_model(seminr_model = CPM_model, nboot = 10000, cores = parallel::detectCores(), seed = 123)																																																							
Output 5.2	Bootstrapping model using seminr... SEMinR Model successfully bootstrapped																																																							
Code 5.3	#SUMMARIZE THE RESULTS OF BOOTSTRAPPING #alpha sets the specified level for significance, i.e., 0.05 sum_boot_CPM_model <- summary(boot_CPM_model, alpha = 0.05)																																																							
Code 5.4	#INSPECT STRUCTURAL PATHS sum_boot_CPM_model\$bootstrapped_paths																																																							
Output 5.4	<table><thead><tr><th></th><th></th><th>Original Est.</th><th>Bootstrap Mean</th><th>Bootstrap SD</th><th>T Stat.</th><th>2.5% CI</th><th>97.5% CI</th></tr></thead><tbody><tr><td>SUPR</td><td>-> ORGC</td><td>0.381</td><td>0.385</td><td>0.070</td><td>5.477</td><td>0.245</td><td>0.516</td></tr><tr><td>SUPR</td><td>-> PERF</td><td>0.064</td><td>0.062</td><td>0.071</td><td>0.901</td><td>-0.077</td><td>0.198</td></tr><tr><td>TEAM</td><td>-> ORGC</td><td>0.464</td><td>0.470</td><td>0.068</td><td>6.788</td><td>0.334</td><td>0.603</td></tr><tr><td>TEAM</td><td>-> PERF</td><td>0.201</td><td>0.201</td><td>0.084</td><td>2.397</td><td>0.037</td><td>0.363</td></tr><tr><td>ORGC</td><td>-> PERF</td><td>0.636</td><td>0.636</td><td>0.080</td><td>7.943</td><td>0.475</td><td>0.788</td></tr></tbody></table>										Original Est.	Bootstrap Mean	Bootstrap SD	T Stat.	2.5% CI	97.5% CI	SUPR	-> ORGC	0.381	0.385	0.070	5.477	0.245	0.516	SUPR	-> PERF	0.064	0.062	0.071	0.901	-0.077	0.198	TEAM	-> ORGC	0.464	0.470	0.068	6.788	0.334	0.603	TEAM	-> PERF	0.201	0.201	0.084	2.397	0.037	0.363	ORGC	-> PERF	0.636	0.636	0.080	7.943	0.475	0.788
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ORGC	-> PERF	0.636	0.636	0.080	7.943	0.475	0.788																																																	
Code 5.5	#INSPECT MODEL R SQUARES summary_CPM_model\$paths																																																							
Output 5.5	<table><thead><tr><th></th><th>ORGC</th><th>PERF</th></tr></thead><tbody><tr><td>R^2</td><td>0.406</td><td>0.619</td></tr><tr><td>AdjR^2</td><td>0.392</td><td>0.606</td></tr><tr><td>SUPR</td><td>0.381</td><td>0.064</td></tr><tr><td>TEAM</td><td>0.464</td><td>0.201</td></tr><tr><td>ORGC</td><td>.</td><td>0.636</td></tr></tbody></table>									ORGC	PERF	R^2	0.406	0.619	AdjR^2	0.392	0.606	SUPR	0.381	0.064	TEAM	0.464	0.201	ORGC	.	0.636																														
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Code 5.6	#INSPECT EFFECT SIZES summary_CPM_model\$fSquare																																																							
Output 5.6	<table><thead><tr><th></th><th>SUPR</th><th>TEAM</th><th>ORGC</th><th>PERF</th></tr></thead><tbody><tr><td>SUPR</td><td>0.000</td><td>0.000</td><td>0.239</td><td>0.008</td></tr><tr><td>TEAM</td><td>0.000</td><td>0.000</td><td>0.357</td><td>0.077</td></tr><tr><td>ORGC</td><td>0.000</td><td>0.000</td><td>0.000</td><td>0.632</td></tr><tr><td>PERF</td><td>0.000</td><td>0.000</td><td>0.000</td><td>0.000</td></tr></tbody></table>									SUPR	TEAM	ORGC	PERF	SUPR	0.000	0.000	0.239	0.008	TEAM	0.000	0.000	0.357	0.077	ORGC	0.000	0.000	0.000	0.632	PERF	0.000	0.000	0.000	0.000																							
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element nested in the `sum_boot_CPM_model` object. We then review the R^2 variance explained (`$paths`) and the f^2 effect sizes (`$fSquare`) for our theoretical model.

From these results, we determine that VIF values are well below the threshold (i.e., $VIF \leq 1.683$) indicating collinearity is not likely a critical problem in the model (Table 5, Output 5.1). Reviewing the size and significance (i.e., based on 10,000 bootstrap subsamples) of the structural paths to ORGC, the antecedents SUPR ($\beta = 0.381$, $t = 5.477$, CI [0.245, 0.516]) and TEAM ($\beta = 0.464$, $t = 6.788$, CI [0.334, 0.603]) each have a moderate, positive impact on organizational commitment. For the outcome of the target construct PERF, ORGC ($\beta = 0.636$, $t = 7.943$, CI [0.475, 0.788]) has the strongest impact, while TEAM ($\beta = 0.201$, $t = 2.397$, CI [0.037, 0.363]) also had a positive but modest impact on PERF. The only nonsignificant path in our model is $SUPR \rightarrow PERF$ ($\beta = 0.064$, $t = 0.901$, CI

[−0.077, 0.198]). The explanatory power for ORGC ($R^2 = 0.406$) and PERF ($R^2 = 0.619$) are at satisfactorily high levels considering the small number of explanators in this simple model, and in comparison to similar studies (e.g., Karem et al., 2019; Khalid et al., 2018). The effect sizes for SUPR $\rightarrow$ ORGC ($f^2 = 0.239$), TEAM $\rightarrow$ ORGC ($f^2 = 0.357$), and ORGC $\rightarrow$ PERF ($f^2 = 0.632$) correspond to the rank order of the path coefficients.

3.6.1 | Structural model—Predictive power

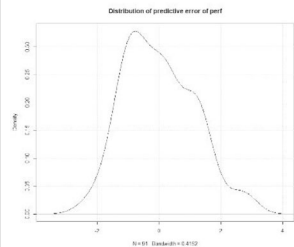
When assessing the predictive power of the structural model in PLS-SEM, researchers focus on the out-of-sample metrics. In PLS-SEM, the PLS_{predict} procedure enables researchers to evaluate a model's out-of-sample predictive power (Shmueli et al., 2016, 2019). To do so, PLS_{predict} uses k -fold cross-validation (CV) to generate separate holdout and test samples from the dataset for computing prediction errors. In comparison to CB-SEM, which provides indeterminate construct scores, PLS-SEM supports this kind of assessment based on its determinate construct scores (see also Rigdon et al., 2017). PLS_{predict} is a recently developed set of procedures which include resampling techniques for the assessment of a given model's predictive ability (Sarstedt, Ringle, et al., 2020; Shmueli et al., 2019). Although PLS_{predict} is a relatively new development within the PLS-SEM framework, predictive algorithm selection based on sample reuse (or resampling) strategies are not new in data analysis, nor are these concepts unique to PLS-SEM (Geisser, 1975; Yarkoni & Westfall, 2017; Yuan et al., 2021). When applying PLS_{predict} for predictive assessment, the procedure is called k -fold CV, where CV refers to the process of “averaging several holdout estimators of the risk corresponding to different data splits” (Arlot & Celisse, 2010, p. 53).

The predictive power assessment compares PLS-SEM's prediction errors with those of a naïve benchmark such as the linear regression model (LM). For an endogenous construct, the LM benchmark regresses each of its indicators on the indicators of all earliest antecedent (EA) constructs or, alternatively, direct antecedent (DA) constructs in the model. When mediating variables are included in the model the EA approach omits the mediator from the assessment, looking only at the predictive power of the exogenous (i.e., earliest) variables in the model, whereas the DA approach considers all antecedents as predictors (Hair et al., 2021, chap. 6). Extant research indicates the DA approach generates predictions with higher accuracy (Danks, 2021), which is highly relevant for research intended to inform forward-looking policies (Liengaard et al., 2021). Therefore, we use the DA approach for our model when assessing the predictive power using the LM benchmark.

Following Shmueli et al. (2016, 2019) and their PLS_{predict} procedure, we anticipate our PLS-SEM results will have higher predictive capabilities than the LM benchmark. The out-of-sample prediction is typically quantified by either the root mean square error (RMSE) or the mean absolute error (MAE). While the RMSE penalizes observations with large errors, the MAE equally includes the error of each observation. The RMSE is typically the default prediction error assessment criterion. The MAE should be used, however, when error distributions are highly nonsymmetrical (i.e., highly skewed; Danks & Ray, 2018; Hair et al., 2021, chap. 6; Shmueli et al., 2019). To determine the appropriate prediction statistic for our model, we visually examine the error distributions using histogram plots. We also review the RMSE (or MAE) results for the key outcome construct as well as the indicators of the mediator construct. For the key target construct in the model, predictive power is determined when the PLS-SEM estimation has lower prediction errors than the LM benchmark (Shmueli et al., 2016, 2019). More specifically, for each indicator of this construct we determine if the prediction error (i.e., RMSE or MAE) is smaller for PLS-SEM model than it is for LM. When comparing all construct indicators, we follow the comparative assessment guidelines presented by Shmueli et al. (2019, pp. 2329–2330):

- PLS-SEM prediction errors < LM for none of the indicators: lack of predictive power.
- PLS-SEM prediction errors < LM for a minority of the indicators: low predictive power.
- PLS-SEM prediction errors < LM for a majority of the indicators: medium predictive power.
- PLS-SEM prediction errors < LM for all indicators: high predictive power.

TABLE 6 Assess predictive power.

Code 6.1	<pre>#GENERATE MODEL PREDICTIONS predict_CPM_model <- predict_pls(model = CPM_model, technique = predict_DA, noFolds = 10, reps = 10)</pre>
Code 6.2	<pre>#SUMMARIZE PREDICTION RESULTS sum_predict_CPM_model <- summary(predict_CPM_model)</pre>
Code 6.3	<pre>#GET PREDICTION RESULTS sum_predict_CPM_model</pre>
Output 6.3	<pre>PLS in-sample metrics: orgc_1 orgc_2 orgc_3 perf RMSE 0.736 0.935 0.746 1.084 MAE 0.592 0.756 0.581 0.899 PLS out-of-sample metrics: orgc_1 orgc_2 orgc_3 perf RMSE 0.754 0.966 0.767 1.131 MAE 0.609 0.786 0.598 0.934 LM in-sample metrics: orgc_1 orgc_2 orgc_3 perf RMSE 0.563 0.677 0.635 0.972 MAE 0.412 0.532 0.532 0.813 LM out-of-sample metrics: orgc_1 orgc_2 orgc_3 perf RMSE 0.637 0.742 0.705 1.109 MAE 0.466 0.582 0.587 0.921</pre>
Code 6.4	<pre>#ANALYZE DISTRIBUTION OF PREDICTION ERROR (PERF) par(mfrow=c(1,1)) plot(sum_predict_CPM_model,indicator = "perf")</pre>
Output 6.4	

When the key target construct only has a single item (i.e., like the target construct PERF in our model), we will conclude the model either has high or no predictive power. Table 6 provides the code and output for conducting predictive power assessment with SEMinR. We run the $PLS_{predict}$ analysis by applying the `predict_pls()` function and specifying the DA approach by assigning `predict_DA` for the technique. Generating the model predictions for the `predict_CPM_model` object will allow us to extract and summarize our error distributions and prediction error values. Based on $PLS_{predict}$ guidance for k -fold CV procedures to estimate a model's ability to predict future observations (Shmueli et al., 2019). When executing $PLS_{predict}$, Shmueli et al. (2016, 2019) suggested setting the value of k to 10 (i.e., 10-fold CV)—see also Liengaard et al. (2021) and Sharma, Liengaard, Hair, et al. (2023) on the number of folds for predictive model assessment. Therefore, we set a value for the number of folds (`noFolds = 10`, `reps = 10`) as shown in Table 6 (Code 6.1). To ensure that our training sample ($k - 1$) meets minimum sample size requirements, we again draw on the inverse square root approach formula presented in Figure 4 (Kock & Hadaya, 2018). For the CPM, setting this value to $k = 10$ results in a training sample of 82 observations, which meets our minimum required sample size of 69 observations when assuming that the minimum path coefficient expected to be significant is

between 0.21 and 0.30 to render the corresponding effect significant at 5% (also see Hair et al., 2022, chap. 1; Manley et al., 2021). For in-depth discussion of $PLS_{predict}$ and the k -fold CV procedure, please see Hair et al. (2021, chap. 6) and Shmueli et al. (2019).

Generating the model predictions for the `predict_CPM_model` object will allow us to extract and summarize our error distributions and prediction error values. As shown in Code 6.2 and 6.3, we use the `summary()` function to create a summary of prediction metrics (`sum_predict_CPM_model`). To determine the appropriate prediction statistic (i.e., RMSE or MAE) for our model, we use the `plot()` function to examine the error distribution of PERF (`indicator = "perf"`) as shown in Code 6.4.

To assess predictive power using the LM benchmark, we first view the distribution error of the PERF construct (Table 6, Output 6.4). We find that PERF's error distribution is relatively symmetrical and therefore use the RMSE metrics in Table 6 (Output 6.3; see PLS out-of-sample metrics: RMSE) to assess predictive power (Danks & Ray, 2018; Shmueli et al., 2019). The single indicator of the target construct (PERF) has a higher RMSE when using PLS-SEM than LM predictions ($RMSE_{PLS-SEM} = 1.131 > RMSE_{LM} = 1.109$). Hence, PLS-SEM cannot beat the LM benchmark and we conclude the model lacks predictive power. In such situations, researchers may choose to evolve their model to improve its predictive power (e.g., by taking into account additional constructs with high predictive relevance and using a larger sample).

As a newer alternative to $PLS_{predict}$, researchers may apply the cross-validated predictive ability test (CVPAT) for PLS-SEM (Liengaard et al., 2021). This test can also be used to evaluate whether the PLS-SEM model results in significantly better out-of-sample predictive accuracy than a naïve benchmark such as the average value prediction or LM (Sharma, Liengaard, Hair, et al., 2023). Since CVPAT is not yet available in the SEMinR package for R, we refer researchers to the R code provided by Sharma, Liengaard, Hair, et al. (2023) for this additional assessment (available from https://github.com/ECONshare/CVPAT_extensions). Finally, to further support causal-predictive PLS-SEM results, researchers may consider recent developments for assessing and reporting model fit (e.g., Guenther et al., 2023; Ringle et al., 2023; Schubert et al., 2023).

3.6.2 | Structural model—Mediation testing

For mediation testing, we follow extant recommendations on PLS-SEM mediation assessment (Hair et al., 2022, chap. 7; Nitzl et al., 2016; Sarstedt, Hair, et al., 2020), which build on mediation guidelines proposed by Zhao et al. (2010). More specifically, this guidance distinguishes between the following mediation results in PLS-SEM:

- *Indirect-only (full) mediation* where the indirect effect via the mediator construct is significant but not the direct effect from the independent to the dependent variable.
- *Complementary (partial) mediation* where both the indirect effect via the mediator construct and the direct effect from the independent to the dependent construct are significant; in addition, the product of direct and indirect effects is positive.
- *Competitive (partial) mediation* where both the indirect effect via the mediator construct and the direct effect from the independent to the dependent construct are significant; in addition, the product of direct and indirect effects is negative.
- *No mediation (with a direct-only or no relationship at all)* where the indirect effect is nonsignificant.

Prior research has suggested the use of the Sobel (1982) test. However, following Preacher and Hayes (2008), current PLS-SEM guidance recommends bootstrapping the sampling distribution which can be more confidently applied to a broader range of sample sizes and yields higher levels of statistical power (e.g., Sarstedt, Hair, et al., 2020). Therefore, we will once again bootstrap our model to determine whether indirect effects are significant at the 5% level for our mediation model. Regarding bootstrapping, we demonstrate the steps as comprehensively as

TABLE 7 Inspect indirect effects.

Code 7.1	#INSPECT TOTAL INDIRECT EFFECTS summary_CPM_model\$total_indirect_effects					
Output 7.1	SUPR TEAM ORGC PERF SUPR 0.000 0.000 0.000 0.243 TEAM 0.000 0.000 0.000 0.295 ORGC 0.000 0.000 0.000 0.000 PERF 0.000 0.000 0.000 0.000					
Code 7.2	#BOOTSTRAP THE MODEL boot_CPM_model <- bootstrap_model(seminr_model = CPM_model, nboot = 10000, cores = parallel::detectCores(), seed = 123)					
Output 7.2	Bootstrapping model using semnr... SEMinR Model successfully bootstrapped					
Code 7.3	#INSPECT SPECIFIC INDIRECT EFFECTS (SUPR -> ORGC -> PERF) specific_effect_significance(boot_CPM_model, from = "SUPR", through = "ORGC", to = "PERF", alpha = 0.05)					
Output 7.3	Original Est.	Bootstrap Mean	Bootstrap SD	T Stat.	2.5% CI	97.5% CI
	0.24257690	0.24524064	0.05688591	4.26427058	0.14092872	0.36211145
Code 7.4	#INSPECT SPECIFIC INDIRECT EFFECTS (TEAM -> ORGC -> PERF) specific_effect_significance(boot_CPM_model, from = "TEAM", through = "ORGC", to = "PERF", alpha = 0.05)					
Output 7.4	Original Est.	Bootstrap Mean	Bootstrap SD	T Stat.	2.5% CI	97.5% CI
	0.29521436	0.29937650	0.05897567	5.00569739	0.18999901	0.41944982

possible, however, if (and only if) no changes have been made to the model, researchers do not need to bootstrap the model again and can instead use the same `summary_CPM_model` object. Next, we assess the significance of bootstrapped direct effects for each antecedent variable on the key performance outcome. Following the steps outlined in Table 7, we first use the function `$total_indirect_effects` to assess the `summary_CPM_model` object to extract total indirect effects (Table 7, Code 7.1). Next, we will bootstrap our model (`bootstrap_model()` and `nboot=10000`; Table 7, Code 7.2) to evaluate each indirect path specified for significance (Table 7, Codes 7.3 and 7.4).

Indirect effects on PERF through ORGC are: $SUPR \rightarrow ORGC \rightarrow PERF$ ($\beta = 0.243$, $t = 4.264$, CI [0.141, 0.362]); and $TEAM \rightarrow ORGC \rightarrow PERF$ ($\beta = 0.295$, $t = 5.006$, CI [0.190, 0.419]). We conclude that the indirect effects are significant at the 5% level, providing general support for our mediation model (Hair et al., 2021, chap. 7, fig. 7.2, p. 142). We will also assess the significance of direct effects for each antecedent variable on the key performance outcome (see Table 8, Code 8.1). The `$bootstrapped_paths` function is then used to extract the direct effects and corresponding CIs for the object `boot_CPM_model`.

Reviewing the results in Table 8 (Output 8.1), we find that the direct effect of ORGC on PERF ($ORGC \rightarrow PERF$) is 0.636 with a 95% CI [0.480, 0.786], indicating this final path is statistically significant. Direct effects for $TEAM \rightarrow PERF$ ($\beta = 0.201$, $t = 2.397$, CI [0.037, 0.363]) are significant at 5% suggesting partial mediation. Direct effects for $SUPR \rightarrow PERF$ ($\beta = 0.064$, $t = 0.901$, CI [-0.077, 0.198]) are nonsignificant at 5% suggesting ORGC fully mediates the impact of SUPR on PERF. Our next step is to determine whether ORGC's role as an intervening variable between TEAM and PERF is complementary or competitive. For the TEAM construct, we will assess whether the product of

TABLE 8 Inspect direct effects.

Code 8.1	#INSPECT CONFIDENCE INTERVALS FOR DIRECT EFFECTS sum_boot_CPM_model\$bootstrapped_paths							
Output 8.1		Original Est.	Bootstrap Mean	Bootstrap SD	T Stat.	2.5% CI	97.5% CI	
	SUPR -> ORGC	0.381	0.385	0.070	5.477	0.245	0.516	
	SUPR -> PERF	0.064	0.062	0.071	0.901	-0.077	0.198	
	TEAM -> ORGC	0.464	0.470	0.068	6.788	0.334	0.603	
	TEAM -> PERF	0.201	0.201	0.084	2.397	0.037	0.363	
	ORGC -> PERF	0.636	0.636	0.080	7.943	0.475	0.788	
Code 8.2	#CALCULATE THE SIGN OF P1*P2*P3 (TEAM) summary_CPM_model\$paths["TEAM", "PERF"] * summary_CPM_model\$paths["TEAM", "ORGC"] * summary_CPM_model\$paths["ORGC", "PERF"]							
Output 8.2	[1] 0.05975809							

the paths (TEAM → PERF × TEAM → ORGC × ORGC → PERF) is positive or negative by multiplying the paths (i.e., P1*P2*P3) as shown in Table 8, Code 8.2. According to Zhao et al. (2010), if the product of indirect and direct effects is negative, competitive mediation is inferred. In contrast, if the product is positive, complementary mediation is inferred (see also suppressor effect/partial mediation, Baron & Kenny, 1986). Results from Table 8 (Output 8.2) reveal the product of the three paths (TEAM → PERF × TEAM → ORGC × ORGC → PERF) is positive (0.060), suggesting that ORGC serves as a complementary (i.e., partial) mediator (Hair et al., 2021, chap. 7; Nitzl et al., 2016; Zhao et al., 2010).

3.7 | Interpreting the results

Results for the final model (illustrative) are presented in Figure 5. Overall, structural model results indicate approximately 60% of the variance in employee performance can be explained by the modeled constructs. Based on mediation analysis, we find that organizational commitment and teamwork each had a direct and positive influence on performance. Alternatively, the influence of supervision on performance occurred through organizational commitment, suggesting organizational commitment serves as a complementary mediator. Based on the strength and direction of paths, as well as indirect effects, we infer that organizational commitment is instrumental to actualizing higher levels of performance. In other words, efforts aimed at performance enhancement could be jeopardized when employees lack a sense of commitment to the success of the company. Furthermore, supervision and teamwork together explained a moderate amount of variance in organizational commitment (i.e., 40%). With the target outcome (i.e., performance) in mind, these results point us to additional considerations (e.g., alternative constructs) which could more effectively promote employees' sense of commitment to the organization and more effectively explain performance outcomes. While our model is purely for illustrative purposes, applied researchers might utilize the results of PLS-SEM analyses similar to this demonstration to support, or even reject, assumptions that drive workforce performance enhancement efforts.

In terms of out-of-sample prediction, our illustrative PLS-SEM model did not outperform the LM benchmark for any of the indicators and therefore demonstrates a lack of out-of-sample predictive capability. In other words, results from the CPM analyses do not lend support for the prediction of new data or future observations in the population (i.e., workforce). While this result may appear discouraging, the line of inquiry produced valuable information. For example, the results may apply only to the recently hired employees and not to the overall population. An HRD professional might apply this evidence to inform discussion with managers regarding current or planned initiatives targeting performance improvement. For example, if a strategy is unlikely to produce similar results across all

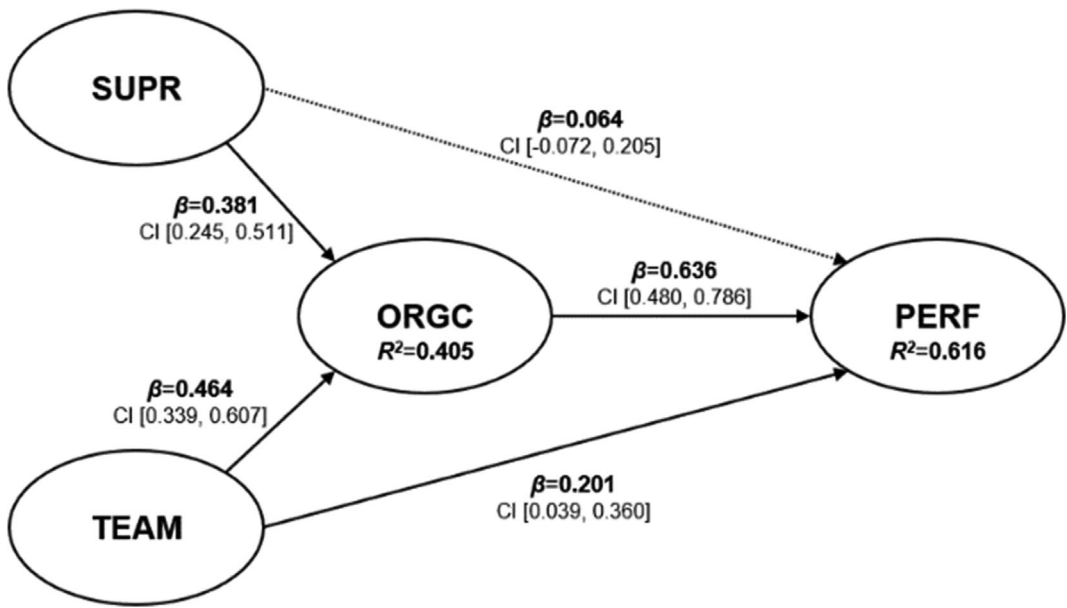


FIGURE 5 PLS-SEM model results. Dotted line (· · · · ·) denotes nonsignificant path ($p > 0.05$). β , path coefficient; CI, confidence interval [lower bound, upper bound]; ORGC, organizational commitment; PERF, performance; SUPR, supervision; TEAM, teamwork.

workforce segments, these analyses can lend support for testing alternative model constellations or employing different research designs (e.g., employee focus groups) to identify possible demographic, cultural, or departmental differences which may influence the impact of performance improvement initiatives.

4 | IMPLICATIONS FOR HRD RESEARCH AND PRACTICE

In this section, we discuss implications of this study for the field of HRD. We first focus on implications for research, followed by the implications for HRD practice.

4.1 | Implications for research

Familiarity with the ever-expanding SEM family of methods differs across and within disciplines and fields, as Bollen et al. (2022, p. 13) noted “some fields are at the forefront, while others have outdated or incorrect knowledge of SEM.” As with all variations and special cases of the SEM framework, the PLS-SEM method has experienced substantial development in recent years and will continue to expand as organizations, and the people who comprise them, evolve. Promoting familiarization with lesser-known methods, including PLS-SEM, can result in a step away from methodological stagnation (Park et al., 2021) and step toward a future at the forefront of research innovation. While our demonstration covers only basic analyses, it is our belief that developing a common understanding of the PLS-SEM methodology will lead to enhanced rigor in areas of research application, consumption, and dissemination.

HRD is a field with strong interdisciplinary foundations (Tkachenko et al., 2017). The demonstration and explanations have implications for members of the HRD community who consume research across related disciplines in which the PLS-SEM method is increasingly common. Understanding the key features of PLS-SEM can promote

development as conscientious consumers of research, especially research occurring in closely connected fields such as human resource management (Ringle et al., 2020) and computer-based learning (Lin et al., 2020). Potential benefits also include opportunities to strengthen collaborative efforts across disciplines and with practitioners seeking new methods for unique organizational contexts.

A final implication of this study for the field involves the methodological justifications used by HRD researchers applying PLS-SEM (e.g., see also Guenther et al., 2023; Rigdon et al., 2017; Ringle et al., 2020, 2023). Surprisingly, the insufficient justifications for the use of PLS-SEM previously discussed (i.e., small samples and relaxed distributional assumptions) were not prominent features in previous HRD studies applying the method. Among the HRD PLS-SEM applications we identified, primary research goals and methodological considerations included: (a) a focus on prediction of endogenous variables (Kunasegaran et al., 2016; Wong et al., 2017); (b) the exploratory nature of the research (Mustafa & Sim, 2023); and (c) the facilitation of conditional mediation (Wong & Roziah, 2015). Note that the possibility to obtain solutions using small samples and non-normally distributed data was a recurring theme in early adopter disciplines. But more recently these disciplines have relied on alternative justifications for choosing PLS-SEM, such as flexibility in handling different measurement model setups including reflective and formative composites, metric, quasi-metric, and categorical scaled data, as well as model complexity (Guenther et al., 2023; Sarstedt, Hair, et al., 2022). Against this backdrop, the HRD studies identified by our review lend optimism for conscientious and rigorous applications of PLS-SEM across the HRD research community.

4.2 | Implications for practice

As Geisser (1975) recognized many years ago, “the prediction of observables or potential observables is of much greater relevance than the estimation of what are often artificial constructs-parameters” (p. 320). The strategic role of HRD professionals is an increasingly important topic in HRD discourse (Gubbins et al., 2018). Developing quantitative competencies across the field of HRD can support the design of interventions and policies directly informed by the workers they are intended to benefit while also addressing salient stakeholder concerns. Use of the current, best evidence to support decision-making in real organizations requires familiarity with a variety of methodological frameworks (Gubbins et al., 2018; Short & Gray, 2018; Tkachenko et al., 2017). Our demonstration is simple, but more importantly, it is one more research tool HRD practitioners will have at their disposal to better explain and understand human behavior in increasingly complex organizational environments. In this regard, the practical value of PLS-SEM lies in the ability to assess context specific interventions and substantiate results which can be inferred to the entire workforce population when theoretical associations are less defined (Hair & Sarstedt, 2021).

Existing software applications support the use of PLS-SEM (Memon et al., 2021; Sarstedt & Cheah, 2019). However, given limited applications of PLS-SEM in the field to draw upon for guidance, practitioners may be hesitant to commit financial resources to subscription-based software at this stage. While the open source R workbook presents both PLS-SEM guidelines and SEMinR usage in a user-friendly manner, it assumes that readers have a basic working knowledge of both R and PLS-SEM (Edeh et al., 2023). Our demonstration was developed to support those with little to no prior experience with PLS-SEM or the R statistical environment. Furthermore, the publicly accessible syntax file can serve as a template which can be adapted for future use to include different constructs and more complex models and analyses.

5 | LIMITATIONS AND FUTURE RESEARCH RECOMMENDATIONS

Our review of PLS-SEM applications in HRD literature suffers from limitations that should be considered. We only included AHRD sponsored journals and *EJTD*, however, recent evidence mapping research indicates a broader range of journals that publish HRD studies (see Pham-Duc et al., 2023) which should be included for a more exhaustive

search. Also, we do not specifically review the HRD PLS-SEM applications identified for misapplication, out-of-date guidelines, or quality of research design. In the spirit of promoting productive dialog surrounding PLS-SEM in the HRD community, highlighting flaws in the application of this emergent methodology was deemed contradictory to our goals for this paper. To further explore this line of inquiry, future research might conduct a more exhaustive search and present a review and critique based on a comprehensive set of records (e.g., see Ringle et al., 2020).

The CPM demonstration presented also has limitations which should be taken into consideration. First and foremost, comparisons and assessments made based on the analyses throughout this paper are for illustrative and educational purposes. All analyses are based on partially simulated data, therefore, results and interpretations are not relevant for empirical comparisons. Second, though our demonstration is based on simulated data, the design includes cross-sectional survey data. Causality cannot be confirmed among observations at a single point in time; to support causal inferences, constructs should be allowed to demonstrate stability over time (e.g., longitudinal design; Maxwell & Cole, 2007). Furthermore, survey data collection methods are subject to a variety of error sources which may provide alternative explanations for findings (i.e., systematic error; Podsakoff et al., 2012). Future contributions might explore procedural and statistical remedies to mitigate and identify sources of common method variance in the context of PLS-SEM (Chin et al., 2013).

Third, we do not demonstrate comparative assessment of theoretically viable alternative models which might offer better out-of-sample predictive efficacy. Future research detailing PLS-SEM model comparison in the R statistical environment and SEMinR (Ray et al., 2022) could be of further use to the HRD community. Recent methodological advancements in PLS-SEM model comparison could support exploration of alternative model configurations that may better explain the mechanisms at play within a given organizational context (Liengaard et al., 2021; Sarstedt, Radomir, et al., 2022; Sharma, Liengaard, Hair, et al., 2023).

Fourth, as brought to our attention by a referee, a limitation of this study is that we did not present readers with a side-by-side comparison of PLS-SEM and CB-SEM. We focus this study on PLS-SEM. For a discussion of PLS-SEM and CB-SEM, we refer the interested reader to the appropriate literature (Cook & Forzani, 2023; Rigdon et al., 2017; Sarstedt et al., 2016). In particular, we recommend the comparison provided by Jöreskog and Wold (1982). At the same time, it should be noted that researchers also consider sum score regression or the generalized structured component analysis for their model estimations. For further details on the method comparisons, we refer the reader to the relevant literature, again (e.g., Cho et al., 2022, 2023; Hair, Hult, et al., 2017; Hair, Matthews, et al., 2017; Marcoulides et al., 2012). Future research, for example, might systematically review the interdisciplinary literature on conflicting PLS-SEM perspectives with special consideration for potential impact on HRD theory and research. Alternatively, journals might encourage submissions which compare and contrast various SEM approaches or conceptually differentiate composite and factor-based models from a measurement theoretic perspective.

Finally, only basic PLS-SEM analyses have been presented, while many more sophisticated techniques would be valuable to the HRD research community (e.g., see Hair et al., 2022, chap. 8). Other techniques in the context of PLS-SEM to explore include the importance–performance map analysis (e.g., Ringle & Sarstedt, 2016), necessary condition analysis (e.g., Richter et al., 2020), higher-order constructs (e.g., Becker et al., 2023), confirmatory tetrad analysis (e.g., Gudergan et al., 2008), endogeneity (e.g., Becker et al., 2022; Hult et al., 2018), heterogeneity in the data (e.g., moderator analysis, multigroup analysis, and latent class segmentation; e.g., Chin & Dibbern, 2010; Memon et al., 2019; Sarstedt, Radomir, et al., 2022), and nonlinear relationships (e.g., Basco et al., 2022).

6 | CONCLUSION

In preparing this methodological article, our main goals were to: (a) introduce basic, but accurate, up-to-date PLS-SEM methodological considerations and assessment guidelines through an easy to follow, transparent, and no-cost to readers demonstration in the R statistical environment; (b) shed light on the efficacy of PLS-SEM as a predictive-

causal modeling approach, which can add diversity to mainstream quantitative methods in the HRD research community; and (c) encourage productive dialog among HRD researchers surrounding leveraging interdisciplinary strengths to contribute to advancing the rigorous application of PLS-SEM to benefit the workforces and organizations that we serve.

As member of the SEM family, PLS-SEM is another valuable analytical tool for HRD researchers exploring prediction-oriented research questions, evaluating complex models, and seeking to identify key factors contributing to intervention outcomes. While our demonstration covers only a basic PLS-SEM application, developing a shared understanding of the method will lead to enhanced rigor in areas of PLS-SEM research application, consumption, and dissemination throughout the HRD community. The demonstrated analyses likewise offer a modest, but important, step in developing additional HRD competencies for quantitative research methods while promoting methodologically rigorous research in the field of HRD.

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DATA AVAILABILITY STATEMENT

All underlying data, software code, and supplementary materials supporting this manuscript are available in the Open Science Framework (OSF) repository: PLS-SEM CPM in R <https://osf.io/jhcx4/>.

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